Florence-2: Advancing a Unified Representation for a Variety of Vision Tasks

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Problem Statement & Challenges

Problem Statement: Existing vision models struggle with handling diverse tasks using a unified framework due to limitations in scalability, adaptability, and comprehensive annotations, requiring a new approach that integrates spatial hierarchy and semantic granularity into a single architecture.

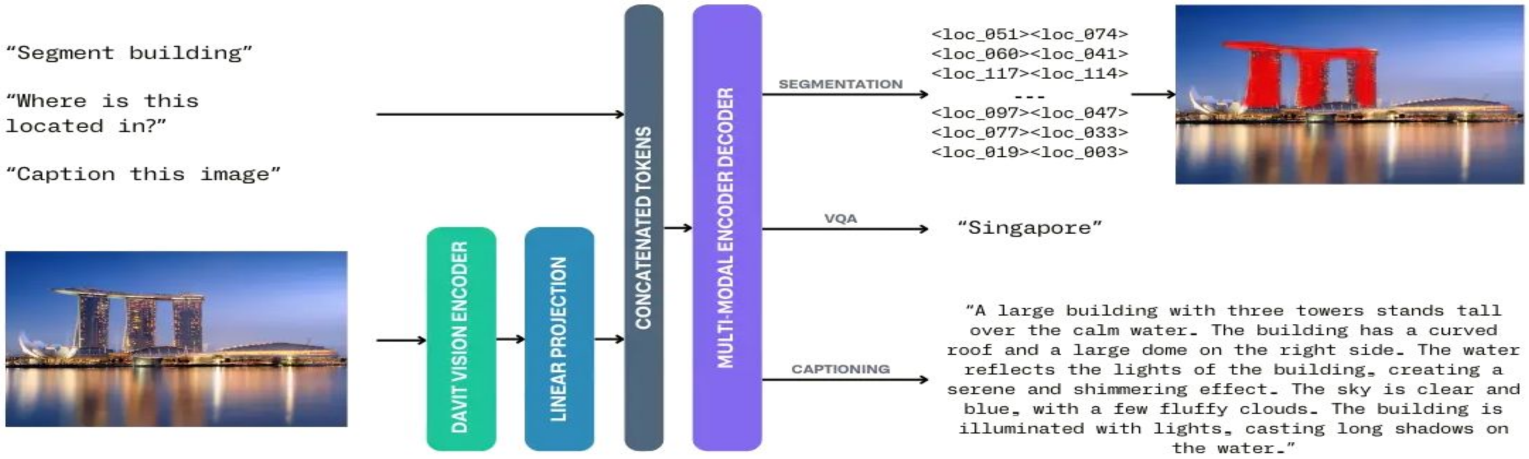
Challenges :

1. Limited scalability & efficiency in handling multiple vision tasks.
2. Task-specific models require independent fine-tuning, leading to increased computational cost.
3. Data limitations: Existing datasets lack comprehensive labeled annotations.
4. No single model integrates spatial hierarchy and semantic granularity effectively.

Florence-2's Solution:

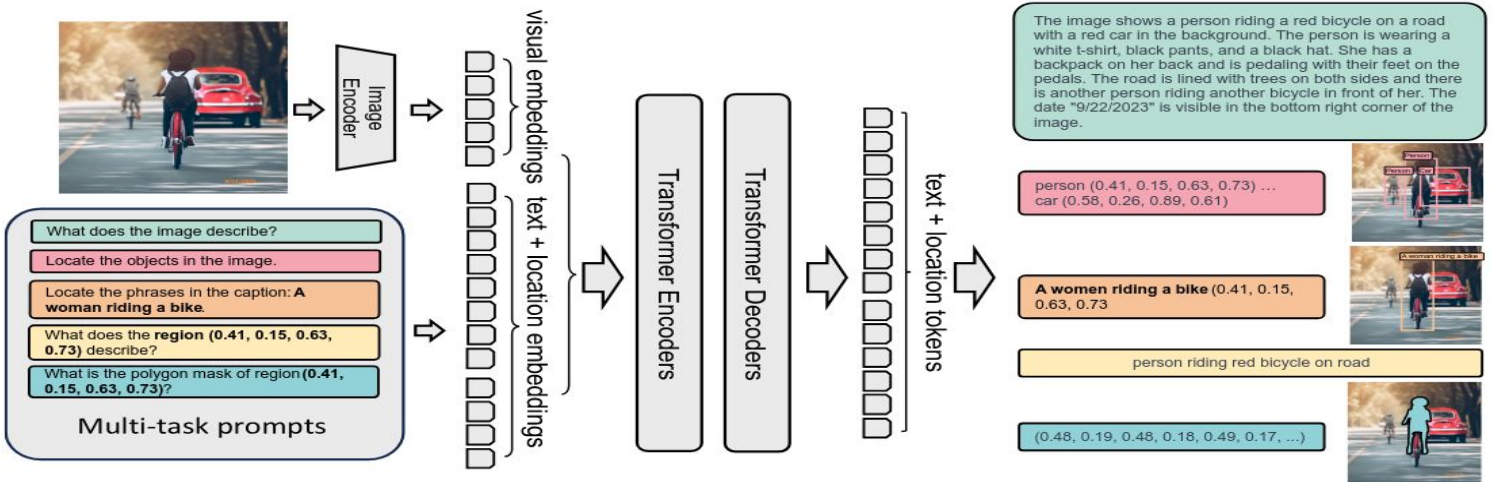
- Provides a unified model for multiple vision tasks.
- Leverages large-scale annotated datasets to improve generalization.
- Uses a sequence-to-sequence learning framework to maintain flexibility across tasks.

Methodology (Florence-2 Approach)



- Unified Vision-Language Model: Handles multiple vision tasks using text prompts instead of separate task-specific architectures.
- Sequence-to-Sequence (seq2seq) Framework:
  1. Uses an image encoder to extract features.
  2. A multi-modality transformer processes both text and image embeddings.
  3. Generates task-specific text-based outputs (e.g., captions, object locations).
- Text-Prompt Adaptation: Enables zero-shot learning across vision tasks without task-specific tuning & Uses extensive pretraining on FLD-5B, making it highly adaptable.

Methodology (Model Architecture)



1. Florence-2 Model Components

Component	Description
Vision Encoder (DaViT – Dual Attention ViT)	Extracts visual token embeddings from images, handling spatial hierarchy efficiently. Weights are initialized from UniCL.
Multi Modality Transformer (Encoder - Decoder)	Processes both text & visual embeddings. Uses cross-attention layers to align vision-language understanding. Weights are initialized from BART.
Text-Based Output Generation	Generates captions, object locations, segment masks, and other results in a text format. Uses a unified loss function for optimization.

Methodology

2. Unified Multi-Task Learning Approach

- FLD-5B dataset annotations are converted into text, making all tasks trainable under a single objective, so No need for separate model heads, tasks are controlled via text prompts.
- Optimization: The model uses a cross-entropy loss function:

$$L = -\sum_{i=1}^{|y|} \log P_\theta(y_i | y_{<i}, x)$$

- Florence-2's seq2seq framework enables task-agnostic adaptability.
- Unified vision-language processing improves scalability and efficiency.

3. Florence-2 Model Variants

Model Size	Image Encoder (DaViT)	Encoder-Decoder	#Params
Florence-2-B	90M params	6 layers, 6 heads	232M
Florence-2-L	360M params	12 layers, 12 heads	771M

Paper Results

Method	#params	COCO Cap. test CIDEr	NoCaps val CIDEr	TextCaps val CIDEr	COCO Det. val2017 mAP
Flamingo	80B	84.3	-	-	-
Florence-2-base	0.23B	133.0	118.7	70.1	34.7
Florence-2-large	0.77B	135.6	120.8	72.8	37.5

zero-shot performance of Florence-2 vision foundation models on image captioning and object detection evaluation tasks.

Experimental Setup & Reproducibility

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Datasets & Metrics

- COCO val2017 – 5 000 images, 80 classes
- PathVQA – 6 000 medical images + Q-A pairs
- COCO Captioning, NoCaps – reproducing similar result was hindered by lack of information (~60% accuracy)

Compute Infrastructure:
SCRUM

- cluster for heavy/long-running jobs (V100/A100, mixed-precision float32)

Local Device

- **CPU**: Intel i7-12700H
- **GPU**: NVIDIA RTX 3060
- **RAM**: 16 GB

```
Using device: cuda:0 Loaded 80 categories
Processing images: 100%|██████████| 5000/5000 [28:19<00:00, 2.94it/s]
Saved 26937 detections to 'coco_detection_results.json'
Total inference time: 0.47 hours
loading annotations into memory... Done (t=0.63s)
creating index... index created!
Loading and preparing results... DONE (t=0.10s)
creating index... index created!
Running per image evaluation... Evaluate annotation type *bbox*
DONE (t=8.42s).
Accumulating evaluation results... DONE (t=1.57s).
Average Precision  (AP) @[ IoU=0.50:0.95 | area=   all | maxDets=100 ] = 0.309
Average Precision  (AP) @[ IoU=0.50      | area=   all | maxDets=100 ] = 0.401
Average Precision  (AP) @[ IoU=0.75      | area=   all | maxDets=100 ] = 0.318
Average Precision  (AP) @[ IoU=0.50:0.95 | area= small | maxDets=100 ] = 0.145
Average Precision  (AP) @[ IoU=0.50:0.95 | area=medium | maxDets=100 ] = 0.309
Average Precision  (AP) @[ IoU=0.50:0.95 | area= large | maxDets=100 ] = 0.441
Average Recall     (AR) @[ IoU=0.50:0.95 | area=   all | maxDets=  1 ] = 0.249
Average Recall     (AR) @[ IoU=0.50:0.95 | area=   all | maxDets= 10 ] = 0.379
Average Recall     (AR) @[ IoU=0.50:0.95 | area=   all | maxDets=100 ] = 0.393
Average Recall     (AR) @[ IoU=0.50:0.95 | area= small | maxDets=100 ] = 0.198
Average Recall     (AR) @[ IoU=0.50:0.95 | area=medium | maxDets=100 ] = 0.416
Average Recall     (AR) @[ IoU=0.50:0.95 | area= large | maxDets=100 ] = 0.561
Evaluation completed.
```

```
---- PathVQA Binary Classification Adversarial Evaluation ----
Total Binary Samples: 3125
Clean Accuracy: 66.3% (2072/3125)
Total evaluation time: 22.74 minutes
```

Limitations of Original Paper

No Adversarial Robustness Evaluation

- The model's vulnerability to FGSM, PGD or any gradient-based attacks is never tested or reported.

Absence of Defense Strategies

- No input or feature-level mitigation techniques are proposed to counter adversarial perturbations.

Lack of Corruption & Noise Benchmarks

- Florence-2 isn't evaluated on corrupted (e.g., ImageNet-C) or noisy variants, leaving its real-world stability unquantified.

No Prompt- or Feature-Perturbation Analysis

- There's no study of how small changes in task prompts or internal features affect output consistency under attack.

Novelty Results

Paper Results	Reproduced Result	Adversarial Attack Impact	Impact Recovery using Defence	Adversarial Type
34.7 mAP	30.9 mAP [IoU 0.5:0.95]	17.8 mAP [IoU 0.5:0.95]	21.2 mAP [IoU 0.5:0.95]	PGD
	40.1 mAP [IoU 0.5]	29 mAP [IoU 0.5]	31.6 mAP [IoU 0.5]	
	31.8 mAP [IoU 0.75]	17.8 mAP [IoU 0.75]	21.6 mAP [IoU 0.75]	
34.7 mAP	30.9 mAP [IoU 0.5:0.95]	24 mAP [IoU 0.5:0.95]	26.2 mAP [IoU 0.5:0.95]	FGSM
	40.1 mAP [IoU 0.5]	34 mAP [IoU 0.5]	36.2 mAP [IoU 0.5]	
	31.8 mAP [IoU 0.75]	25.2 mAP [IoU 0.75]	28 mAP [IoU 0.75]	

This table summarizes end-to-end results on the **COCO object detection dataset**, from clean performance to adversarial attack impact and defense recovery for the Florence-2 base model.

Despite strong PGD and FGSM attacks, our defense consistently recovers 3–8 mAP across IoU thresholds, proving robust recovery under adversarial settings.

Proposed Novelty

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Adversarial Attacks on Florence-2:

- **FGSM** (Fast Gradient Sign Method):
  - Single-step perturbation along gradient sign.(ε)
  - Fast computation; tests model's immediate sensitivity to high-frequency noise

- **PGD** (Projected Gradient Descent):
  - Iterative, multi-step (α, n iters) within ε ball
  - Stronger, finds worst-case perturbations; slower runtime

Why Both?

- FGSM probes initial vulnerability "surface"
- PGD stresses the model against worst-case iterative attacks

Layered Defense Framework:

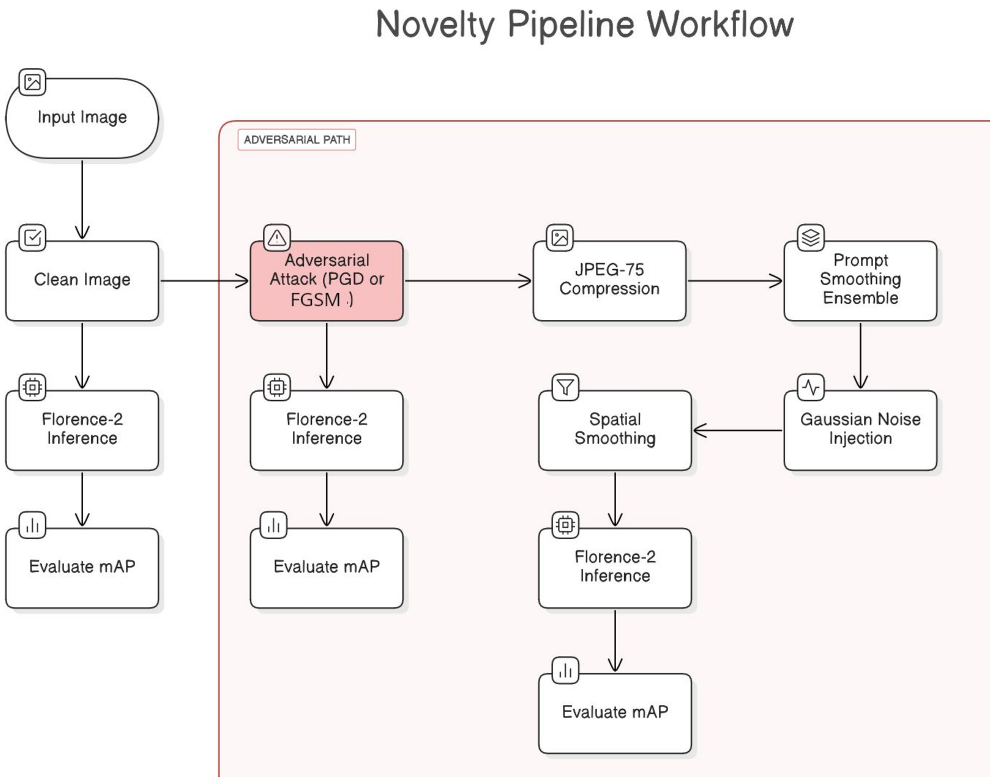
- **External Defenses** (Prompt Smoothing Ensemble)
- **Internal Defenses** (Gaussian Noise Injection, Spatial Smoothing)
- Conditional Integration via single **run_detection** pipeline

Proposed Novelty

Defense Layers & Pipeline Mechanics:

- **External (Input-Level) Layers:**
    - **JPEG-75 Compression:** Disrupts adversarial high-freq artifacts
    - **Prompt Smoothing Ensemble**: 4 variant prompts & minor image augmentations

- **Internal (Feature-Level) Layers:**
    - **Gaussian Noise Injection**: Masks small perturbations in tensor space
    - **Spatial Smoothing**: Blurs residual high-frequency adversarial noise



Adversarial Attacks

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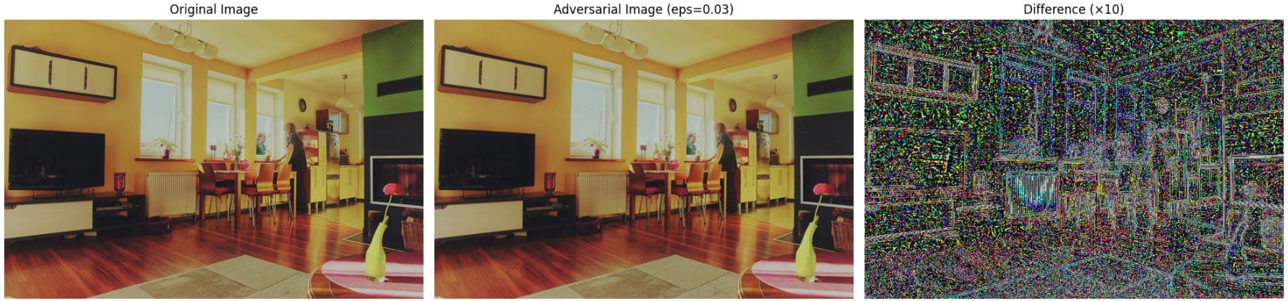
FGSM (Fast Gradient Sign Method):

- Single-step ε perturbation applied along the sign of the loss gradient
- Produces high-frequency, edge-like noise patterns



PGD (Projected Gradient Descent):

- Iterative multi-step attack (α (0.05) steps within an ε-ball)
- Generates subtler, more uniformly distributed perturbations

Novelty Output

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Consistent Recovery Across Data Sizes: Whether using half or full COCO, the absolute AP gain from defense remains comparable showing our framework generalizes to different dataset volumes.

Scalable to Attack Strength: Stronger attacks (ε values) incur larger AP drops but also yield larger absolute recoveries, demonstrating our defenses scale with attack severity.

Higher Relative Recovery on FGSM: Although FGSM induces smaller AP drops, our defenses reclaim a larger *fraction* of that loss compared to PGD, highlighting where our methods are most effective.

```
---- PGD Defense Effectiveness Analysis ----
Clean AP: 0.1576
Adversarial AP (PGD, eps=0.03): 0.0892
Defended AP (PGD): 0.1058
Performance drop due to PGD attack: 0.0684 AP (43.4%)
Recovery by defense: 0.0167 AP (24.4% of drop recovered)

Total PGD evaluation time: 20781.12 seconds (346.35 minutes)
```

Half-Dataset, ε=0.03 (PGD alpha=0.005, iter=10)

```
---- PGD Defense Effectiveness Analysis ----
Clean AP: 0.295
Adversarial AP (PGD, eps=0.04): 0.152
Defended AP (PGD): 0.186
Performance drop due to PGD attack: 0.143 AP (48.6%)
Recovery by defense: 0.034 AP (23.8% of drop recovered)

Total PGD evaluation time: 50764.21 seconds (846.07 minutes)
```

Full-Dataset, ε=0.04 (PGD alpha=0.005, iter=10)

FGSM

```
---- Defense Effectiveness Analysis ----
Clean AP: 0.1576
Adversarial AP: 0.1199
Defended AP: 0.1315
Performance drop due to attack: 0.0376 AP (23.9%)
Recovery by defense: 0.0115 AP (30.6% of drop recovered)

Total FGSM evaluation time: 15972.98 seconds (266.22 minutes)
```

FGSM

```
---- Defense Effectiveness Analysis ----
Clean AP: 0.295
Adversarial AP: 0.233
Defended AP: 0.261
Performance drop due to attack: 0.0619 AP (21.0%)
Recovery by defense: 0.0289 AP (46.7% of drop recovered)
Total FGSM evaluation time: 30996 seconds (516.6 minutes)
```

Impact of Novelty

Synergistic Defense Gains

- Combined input-level & feature-level layers help to easily recover from adversarial data
- Defense recovers up to ~50% of FGSM loss and ~30% of PGD loss

Practical Robustness Boost

- Balances compute cost vs. security gain, making it ready for real-world deployment against adversarial threats.

Modular, Flag-Driven Pipeline

- Single pipeline allows on/off toggling of each defense.
- Enables rapid ablation studies and future extension

Model-Agnostic & Efficient

- Plug-and-play with any transformer-based vision model
- Runs entirely in float16 on GPU for minimal extra compute

Potential Drawback

- Increased end-to-end inference time
- Minor clean-image mAP drop (~1–2%) due to aggressive smoothing/noise

Challenges Faced

Incomplete Model Specs:

- No hyperparameter or configuration details for Florence-2 in the paper or online
- Required building inference/training code from scratch

Inconsistent Zero-Shot Scores:

- COCO captioning & “no cap” caption tasks failed to match reported performance
- 20–30 full runs & parameter tweaks still couldn’t replicate published mAP/CIDEr

Compute & Throughput Constraints:

- Used batch size 1 to prioritize accuracy & it impacted on inference throughput.
- Full end-to-end COCO object detection run (val set) took ~8–10 hours on GPU

Trial-and-Error Workflow:

- Extensive manual tuning of ε, α, prompt variants and defense flags
- Slow iteration cycle & each adjustment required a full multi-hour run

Documentation Gaps:

- Sparse guidance on task-prompt engineering and internal parameters
- Debugging and reproducing model internals proved highly time-consuming

Conclusion

- Demonstrated that Florence-2 Base is vulnerable to standard adversarial attacks in a zero-shot setting, emphasizing the need for robustness evaluation.
- Introduced a multi-layer defense framework, combining lightweight input transforms and in model feature defenses that substantially restores detection performance with minimal overhead.
- Delivered a modular, flag-driven pipeline for easy toggling, ablation studies, and seamless integration of future defenses.

Future Work

Improve Reproducibility Across Datasets

- Continue experimenting on other datasets (e.g., VQA, captioning, grounding) to align results with baselines and evaluate generalization.

Optimize the Defense Framework

- Refine the current layered setup to reduce runtime overhead and improve efficiency, especially for real-time or large-scale deployment scenarios.

Explore Alternative Defense Mechanisms

- Investigate deeper integration methods like model hooks, adaptive feature-level defenses, as current internal trials with hooks were unsuccessful.

Enhance Flexibility & Automation

- Develop a more automated pipeline for batch processing, parameter tuning, and result logging to streamline future experimentation and scaling.

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Thank you